



Airline



VOYA

~A COLLABORATIVE TRAVEL PLANNING APPLICATION

Presented By: Sanyukta Bhadlikar, Justin
Castro, Oluwawemimo Jayeoba, Albert
Jimenez, Amrr Okasheh, Anish Patel &
Suhani Rana

BOARDING PASS

- **GROUP**

Team 8

- **COURSE**
CS/SE/CE
3354

- **SEMESTER**
Fall '25





Airline



BOARDING PASS

OUR OBJECTIVE



Airline



BOARDING PASS

Objective

- Centralize group travel planning
- Support real time collaboration
- Simplify budgeting
- Make finding and choosing activities easier
- Improve communication
- Enable live location sharing for group safety



Airline



BOARDING PASS

COST ESTIMATION



Hardware & Software Licensing Costs



Hardware:

- iPhone + Android test devices: \$600-800
- Cloud servers, Database hosting, and storage(~3 months): \$150
- Backup hardware: \$100
- Total Hardware: \$850-1050

Software & Licensing:

- Development tools: Free
- APIs(Google Maps, Email, Firebase Cloud Messaging): \$20-30
- Optional cloud test devices: \$20-30
- Total Software: \$30-50



Personnel, Training, & Total Cost

Personnel:

- Student team of 7
- Symbolic academic rate: \$10 per hour
- Estimated effort: 544 hours
- Personnel Cost: \$5,400
- Industry equivalent (\$25-35/h): \$13,600-19,040

Training costs:

- Training would utilize mostly free online resources: \$0-30

Total Development Cost:

- Total Project Cost: \$6,280-6,530





Airline



BOARDING PASS

PROJECT TIMELINE



Airline



Project Timeline 2026

Timeline 20 Jan 10 Feb 3 Mar 25 Mar 8 Apr 20 Apr

Increment 1:

- Setup
- Auth
- Architecture

Increment 2:

- Trip Management
- Services

Increment 3:

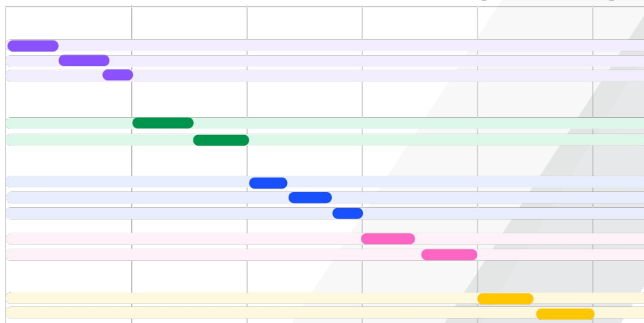
- Budgeting
- Polls
- Reports

Increment 4:

- API Integration
- Scheduler

Increment 5:

- Final Integration
- Final Testing



INCREMENT 1 - SETUP

Architecture, setup, authentication

INCREMENT 2 - TRIPS

Trip creation, viewing, and member management

INCREMENT 3 - BUDGETS/POLLS

Budget tracking, expenses, polls, and reports

INCREMENT 4 - INTEGRATION

Maps, location sharing, notifications



Airline



BOARDING PASS

FUNCTIONAL REQS



Airline



Functional Specifications

| AUTHENTICATION

Allows users to sign in with email and password and validates their credentials.

| TRIP MANAGEMENT

Admins can create, edit, delete trips and manage trip members.

| USER TRIP ACCESS

Users can view trips they own or are part of, including titles and dates.

| LOCATION TRACKING

Shows the user's current location and the last known location and members.

| BUDGETING

Users can add and delete budget entries, calculate totals, and view remaining budget.

| POLLS & NOTIFICATIONS

Users can create polls, vote, view results, and receive trip related alerts.



Airline



BOARDING PASS

**NON
FUNCTIONAL
REQS**



| Usability

Makes the app easy to learn with onboarding, tips, and intuitive first-time use.

| Performance

Keeps the app fast, responsive, and able to handle many users at once.

| Space

Keeps the app small on devices and limits how much data each trip stores.

| Dependability

Shows helpful error messages and keeps important trip and budget data backed up.

| Security

Uses secure login, encrypted communication, and protects trip information.

| Environmental

Works on both iOS and Android and supports light and dark mode.



Airline



| Operational

Runs 24/7, handles maintenance during low traffic, and fits common phone screen sizes.

| Developmental

Uses standard tools and GitHub so the whole team can collaborate smoothly.

| Regulatory

Lets users delete their accounts and removes all personal data when they do.

| Ethical

Collects and shares data only with consent and clearly explains data use and storage.

| Accounting

Tracks expenses, budgets, timestamps, and who created each entry.

| Safety & Security

Never share a user's location without explicit permission.



Airline



BOARDING PASS

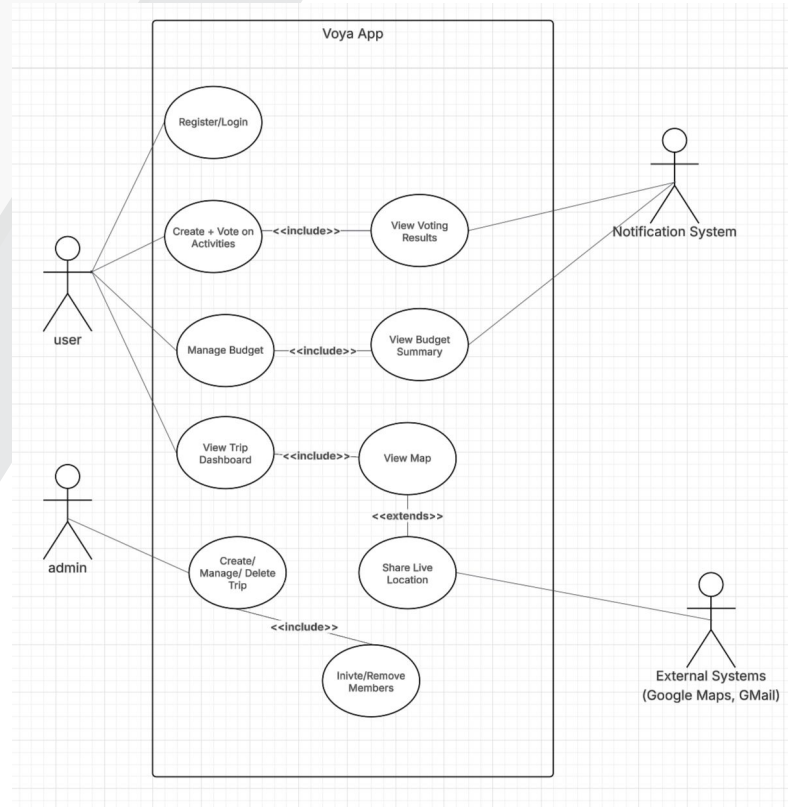
USE CASE DIAGRAM



Airline



Use case Diagram





Airline



SEQUENCE DIAGRAM



CREATING A TRIP

BOARDING PASS

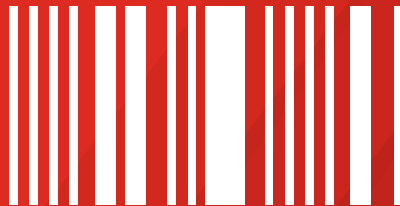
• LIFELINES
CLIENTAPP



TRIPSERVICE



TRIPDATABASE

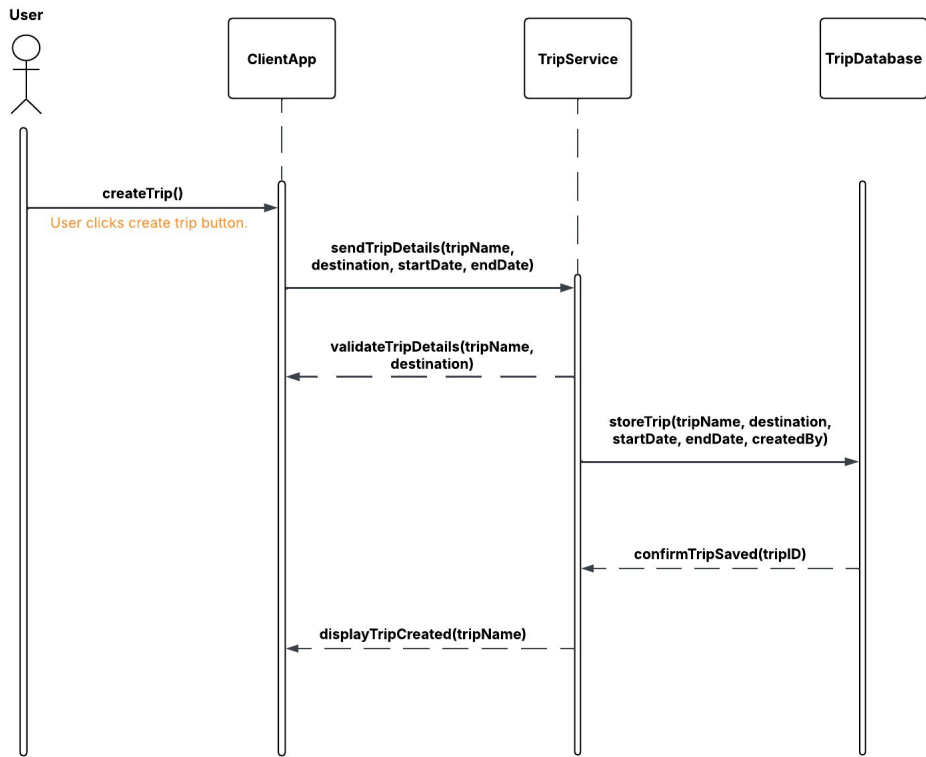




Airline



BOARDING PASS





KEY SYSTEM BEHAVIORS

- **Client** → **Server** → **Database** interaction
- **Backend validation** ensures proper trip setup
- **Database persists trip name, dates, and admin ID**
- **Synchronous calls** provide immediate user feedback
- **Ensures reliability** before adding members or activities



Airline



BOARDING PASS

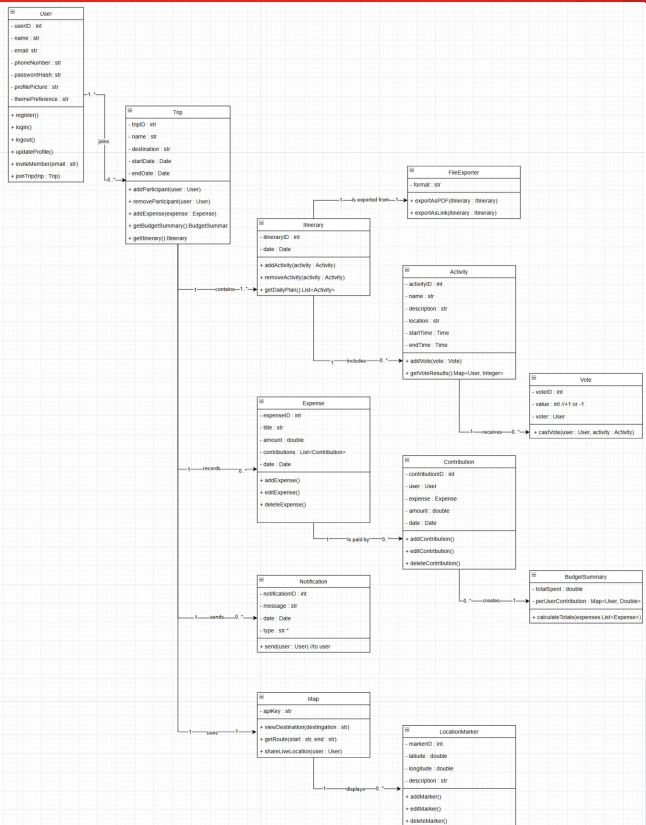
CLASS DIAGRAM



Airline



BOARDING PASS



- Centered around User and Trip class
- Branches off into multiple subsystems
- Makes use of multiple specialized classes



Airline



BOARDING PASS

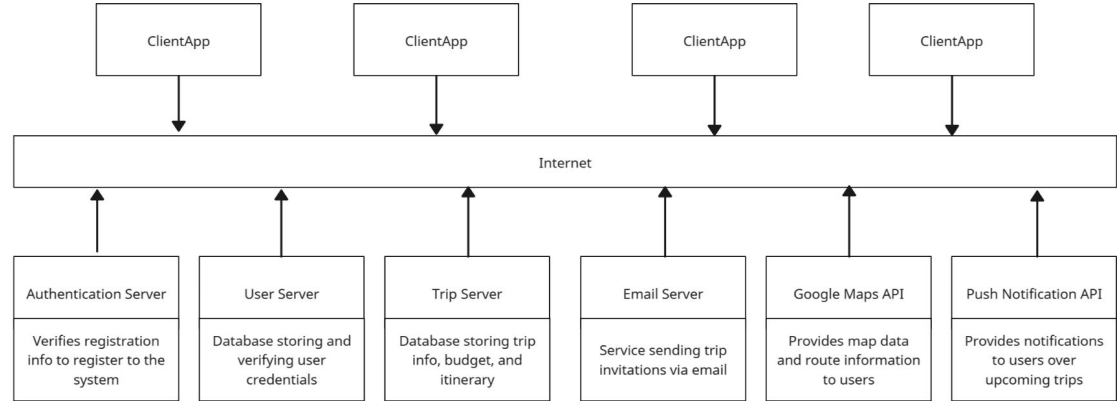
ARCHITECTURE DESIGN



Airline



Client-Server Architecture





Airline



BOARDING PASS

VOYA VS. SIMILAR SYSTEMS



VOYA'S DIFFERENTIATORS



Airline



COMPARISON WITH EXISTING TRAVEL APPS

Context:

We compared Voya with four popular platforms:

Tripadvisor, Triplt, Wanderlog, and Wego

★ *Only Voya offers **complete** group-travel coordination features!*



Tripadvisor & Wego

Strong discovery/booking



Triplt

Excellent personal itinerary generation



Wanderlog

Collaborative planning but limited budgeting + no polls



Airline



Comparative Feature	Voya	Wanderlog	Tripadvisor	Wego	TripIt
Group Itinerary	✓	✓	✗	✗	✗
Shared Budgeting	✓	Limited	✗	✗	✗
Activity Voting	✓	✗	✗	✗	✗
Live Location	✓	✗	✗	✗	✗
Real-Time Notifications	✓	✓	✗	✗	Limited
Map Planning	✓	✓	Limited	✗	✓
Travel Booking	Limited	✗	Limited	✓	✗
Best for Groups	✓	✓	✗	✗	✗



Airline



BOARDING PASS

JUNIT TESTING



Airline



WHAT ARE WE TESTING?

```
BudgetSummary  
- totalSpent : double  
- perUserContribution : Map<User, Double>  
+ calculateTotals(expenses: List<Expense>)
```

UNIT I TESTED

Budget Summary



```
- totalSpent : double  
- perUserContribution : Map<User, Double>
```

calculateTotals()



```
+ calculateTotals(expenses:List<Expense>)
```

Total Spent & per User Contribution





Airline



TEST CASE DESIGN

Expenses

```
Expense food = new Expense( expenseID: 1, amount: 800.0,  
Expense Ubers = new Expense( expenseID: 2, amount: 200.0,  
Expense hotel = new Expense( expenseID: 3, amount: 1000.0,
```

```
userID: "Amrr", amount: 100);  
userID: "Albert", amount: 200);  
userID: "Sanyukta", amount: 300);  
userID: "Suhani", amount: 400);  
userID: "Joyce", amount: 500);|  
userID: "Justin", amount: 600);  
userID: "Anish", amount: 700);
```

Contributions



Airline



JUnit Code Snippet

```
BudgetSummary summary = BudgetSummary.calculateTotals(List.of(food, Ubers, hotel));  
assertEquals( expected: 2000.0, summary.getTotalSpent());  
assertEquals( expected: 7, summary.getPerUserContribution().size());
```

```
assertEquals( expected: 100.0, summary.getPerUserContribution().get("Amrr"));  
assertEquals( expected: 200.0, summary.getPerUserContribution().get("Albert"));  
assertEquals( expected: 300.0, summary.getPerUserContribution().get("Sanyukta"));  
assertEquals( expected: 400.0, summary.getPerUserContribution().get("Suhani"));  
assertEquals( expected: 500.0, summary.getPerUserContribution().get("Joyce"));  
assertEquals( expected: 600.0, summary.getPerUserContribution().get("Justin"));  
assertEquals( expected: 700.0, summary.getPerUserContribution().get("Anish"));
```



Airline



BOARDING PASS

THANK YOU!

CREDITS: This presentation template was created by Slidesgo, and includes icons by Flaticon and infographics & images by Freepik